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Math 455 notes

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Abstract

Personal notes for math 455

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Theorem 1. Let (X, d) be a metric space, then TFAE :

1. X is compact
2. X is sequentially compact, any sequence has a convergent subsequence
3. X is complete and totally bounded, $\forall \varepsilon > 0$, X can be covered by finitely many balls of radius ε

Proof. See Royden and Fitzpatrick □

Proposition 2. If X is compact then $(C(X), \|\cdot\|_{\max})$ is complete

Definition 3 (equicontinuous). $\mathcal{F}$ is a collection of functions from X to $\mathbb{R}$. $\mathcal{F}$ is called equicontinuous at $x \in X$, if $\forall \varepsilon > 0$, $\exists \delta > 0$ such that if $d(x, y) < \delta$, then $|f(x) - f(y)| < \varepsilon$, for all $f \in \mathcal{F}$.

Remark 4. If $\mathcal{F} = \{f_1, f_2, \dots, f_n\}$ finite set, then it's equicontinuous.

Definition 5 (pointwise bounded). The set $\{f_n : X \rightarrow \mathbb{R}\}_{n \in \mathbb{N}}$ is pointwise bounded iff $\forall x \in X$, $\{f_n(x) : n \in \mathbb{N}\}$ is a bounded subset in $\mathbb{R}$

Definition 6 (Uniformly bounded). $\{f_n\}_{n \in \mathbb{N}}$ is uniformly bounded iff $\exists M > 0$ such that $|f_n(x)| \leq M \forall x \in X, \forall n \in \mathbb{N}$

Lemma 7 (Arzale-Ascoli). Let X be a separable metric space, $\{f_n\}_n$ an **equicontinuous sequence** in $C(X)$ that is **pointwise bounded**. Then a subsequence of $\{f_n\}$ converges pointwise on all X to a function on X (not necessarily continuous).

Proof. Let $\{x_j\}_{j=1}^\infty$ be a countable dense subset of X . Look first at the set $\{f_n(x_1)\}_{n \in \mathbb{N}}$, from pointwise boundedness, this is a bounded subset of the reals. From from that, there exists a subsequence, denoting the indices to be $S(1, n) \subseteq \mathbb{N}$, so $\exists \{f_{S(1, n)}(x_1)\}_{n \geq 1}$ where $f_{S(1, n)}(x_1) \rightarrow y_1$ (why? Recall the Bolzano-Weierstrass theorem from analysis 1: every bounded sequence in the reals has a convergent subsequence), let g be such that $g(x_1) = y_1$. Furthermore, there exists a bounded sub-subsequence where $\{S(2, n)\}_n \subseteq \{S(1, n)\}_n$, then let $f_{S(2, n)}(x_2) \rightarrow y_2$ and let $g(x_2) = y_2$. The induction step is that given a $k \in \mathbb{N}$, choose $S(k-1, n) \subseteq S(k-2, n) \subseteq \dots \subseteq S(1, n)$ such that for all $j < k$, $f_{S(j, n)}(x_j) \rightarrow y_j = g(x_j)$. Then let $n_k = S(k, k)$, this is the diagonal subsequence. Letting $j \in \mathbb{N}$, then $\{n_k\}_{k=j}^\infty$ is a subset of $S(k, n)$ such that $f_{n_k}(x_j) \rightarrow y_j = g(x_j)$ (sequence converges then subsequence converges). Let $D = \{x_j\}_{j=1}^\infty$ be a dense subset of X . Then $\{f_{n_k}\}_{k=1}^\infty$ converges pointwise to g on D . Let $x_0 \in X$. Then we have a claim: $\{f_{n_k}(x_0)\}_{k \in \mathbb{N}}$ is cauchy in $\mathbb{R}$. The proof is as follows: Let $\varepsilon > 0$. We know that $\{f_n\}$ is equicontinuous at x , so $\exists \delta > 0$ such that if $d(x, x_0) < \delta$, $|f_n(x) - f_n(x_0)| < \frac{\varepsilon}{3} \quad \forall n \in \mathbb{N}$. Since D is dense in X , $\exists x \in D$ such that $d(x_0, x) < \delta$. Since $x \in D$, then $f_{n_k}(x) \rightarrow g(x)$ as $k \rightarrow \infty$. $\exists N$ such that $\forall n_{k_1}, n_{k_2} > N$, $|f_{n_{k_1}}(x) - f_{n_{k_2}}(x)| < \frac{\varepsilon}{3}$. Then, by the triangle inequality, letting δ be as before,

$$\begin{aligned} |f_{n_{k_1}}(x_0) - f_{n_{k_2}}(x_0)| &\leq |f_{n_{k_1}}(x_0) - f_{n_{k_1}}(x)| + |f_{n_{k_2}}(x) - f_{n_{k_2}}(x_0)| + |f_{n_{k_1}}(x) - f_{n_{k_2}}(x)| \\ &\leq 3 \cdot \frac{\varepsilon}{3} \end{aligned}$$

so $f_{n_k}(x_0) \rightarrow g(x_0)$ as $k \rightarrow \infty$, for all $x_0 \in X$, thus verifying the claim and showing that $\{f_{n_k}\}_{k \in \mathbb{N}}$ converges pointwise to g , so there always exists a subsequence to converges pointwise to some function that's not necessarily continuous. $\square$

Definition 8 (uniformly equicontinuous). $\forall x, y \in X, \varepsilon > 0, \exists \delta$ not depending on x nor y such that if $d(x, y) < \delta$, then $|f(x) - f(y)| < \varepsilon$. This is what you expected knowing the definition of equicontinuous (nonuniform).

Theorem 9 (Arzale-Ascoli). Let X be a compact metric space, let $\{f_n\}_{n \in \mathbb{N}}$ be a uniformly bounded and equicontinuous sequence of functions $f_n: X \rightarrow \mathbb{R}$, then $\exists f_{n_k}$ (subsequence) such that f_{n_k} converges uniformly on X to $f(x) \in C(X)$.

Remark: This is different from the Arzale-Ascoli lemma where X is not required to be compact, $\{f_n\}_{n \in \mathbb{N}}$ needed only to be pointwise bounded and equicontinuous, such that the function representing the convergent limit needn't be continuous.

Proof. X compact implies X is separable. By A-A lemma, $\exists f_{n_k}$ such that $f_{n_k}(x) \rightarrow f(x) \quad \forall x \in X$. To simplify notation let $f_{n_k} = f_n$ for all $k \in \mathbb{N}$. **Remark:** if X compact, $\{f_n\}$ equicontinuous implies $\{f_n\}$ is uniformly equicontinuous $\forall n \in \mathbb{N}$.

Then, knowing that $\{f_n(x)\}$ is a Cauchy sequence in $\mathbb{R}$ for all $x \in X$, let $\varepsilon > 0$, we know that there exists a $\delta > 0$ such that $\forall x, y \in X$ if $d(x, y) < \delta$ then

$|f_n(x) - f_n(y)| < \frac{\varepsilon}{3} \quad \forall n \in \mathbb{N}$. Since X is compact, then X is totally bounded, that means that there exists some finite collection of points $\{x_1, \dots, x_m\}$ such that

$$\bigcup_{i=1}^m B(x_i, \delta) \supseteq X$$

where $f_n(x_i)$ is a Cauchy sequence for all $1 \leq i \leq m$. Then $\exists N \in \mathbb{N}$ such that $\forall n_1, n_2 > N, \forall 1 \leq i \leq m, |f_{n_1}(x_i) - f_{n_2}(x_i)| < \frac{\varepsilon}{3}$. Let $x \in X$, then for some $1 \leq j \leq m, d(x, x_j) < \delta$. Then

$$\begin{aligned} |f_{n_1}(x) - f_{n_2}(x)| &\leq |f_{n_1}(x) - f_{n_1}(x_j)| \\ &\quad + |f_{n_1}(x_j) - f_{n_2}(x_j)| \\ &\quad + |f_{n_2}(x_j) - f_{n_2}(x)| \\ &\leq 3 \cdot \frac{\varepsilon}{3} \end{aligned}$$

Thus $\{f_n\}$ is uniformly Cauchy, thus Cauchy with respect to the max norm, thus uniform convergent, since the limit of continuous functions on $C(X)$ where X is compact is continuous (PS3). $\square$

Theorem 10 (Compactness in $C(X)$). Let X compact metric space, $\mathcal{F} = \{f_\alpha\}_{\alpha \in I} \subseteq C(X)$. $[f_\alpha: X \rightarrow \mathbb{R}]$. Then $\mathcal{F}$ is compact as a subspace of $C(X)$. If $d_\infty(f, g) = \sup_{x \in X} |f(x) - g(x)| = \max_{x \in X} |f(x) - g(x)|$ since X compact is maximum is attained $\iff \mathcal{F}$ is closed, uniformly bounded and equicontinuous.

Remark 11. Let $1 \leq p \leq \infty$, let $S \subseteq l_p = \{x = (x_1, x_2, \dots, x_n, \dots): \sum_{i=1}^\infty |x_i|^p < +\infty\}$ such that S is closed and bounded, then S is compact if and only if S is uniformly equisummable, that is $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\sum_{k=N}^\infty |x_k| < \varepsilon \quad \forall x \in (x_1, \dots, x_n, \dots) \in S$.

Definition 12 (interior and exterior). Let $A \subseteq (X, d)$. We define $Int(A) = \{x \in A: \exists r > 0 \text{ s.t } B(x, r) \subseteq A\}$ and $Ext(A) = Int(A^c)$. We have that $X = Int(A) \sqcup Ext(A) \sqcup Bd(A)$.

Definition 13 (dense). $A \subseteq X$ is dense if every open subset of X contains a point in A

Definition 14 (hollow). $B \subseteq X$ is hollow if $Int(B) = \emptyset$

claim: B is hollow iff B^c is dense in X

Theorem 15 (Baire Category Theorem). The theorem states many things

- Let $\{U_n\}_{n=1}^\infty$ be a countable collection of open and dense subsets of X , then $\bigcap_{n=1}^\infty U_n$ is dense in X
- If $\{V_n\}_{n \in \mathbb{N}}$ is a countable collection of closed and hollow subsets of X , then $\bigcup_{n=1}^\infty V_n$ is hollow in X .

Lemma 16. Let X be a complete m.s. Let $\overline{B(x_1, r_1)} \supseteq \overline{B(x_2, r_2)} \supseteq \dots$ such that $r_j \rightarrow 0$ as $j \rightarrow \infty$. Then $\bigcap_{n=1}^{\infty} B(x_n, r_n) \neq \emptyset$

Proof. Center of balls form a Cauchy sequence □

Definition 17 (nowhere dense). $E \subseteq X$ is nowhere dense if $\bar{E}$ is hollow (has $\emptyset$ interior).

Corollary 18 (BCT C1). Let X be a complete m.s, $\{F_n\}_{n \in \mathbb{N}}$ be a collection of closed sets. Suppose $\bigcup_{n=1}^{\infty} F_n$ has non empty interior. Then at least one of F_n has a non empty interior.

Corollary 19 (BCT C2). Let X be a complete m.s, $\{F_n\}_{n \in \mathbb{N}}$ collection of closed sets, then $\bigcup_{n=1}^{\infty} Bd(F_n)$ is hollow.

Lemma 20. A closed implies $Bd(A)$ hollow in X (exercise)

Theorem 21 (Uniformly bounded restricted to open set). Let $\mathcal{F}$ be a family of continuous functions $f: X \rightarrow \mathbb{R}$. Suppose $\mathcal{F}$ pointwise bounded, that is $\forall x \in X$, $\exists M_x > 0$ such that $\forall f \in \mathcal{F} |f(x)| \leq M_x$. Then there exists a non empty open $U \subseteq X$ such that $\exists M > 0$ where $\forall x \in U, \forall f \in \mathcal{F}, |f(x)| \leq M$

Theorem 22. Let X be a complete metric space, $f_n: X \rightarrow \mathbb{R}$ continuous $\forall n$, $f_n(x) \rightarrow f(x)$, $\forall x \in X$ as $n \rightarrow \infty$ (i.e) $\{f_n\}$ converges pointwise. Then $\exists D$ dense subsets such that $\{f_n\}$ is equicontinuous on D and $f(x)$ is continuous $\forall x \in D$.

We introduce an extra set of notation, $A \subseteq X$ is of the 1st Baire category if A is a union of a countable collection of nowhere dense subsets of X . If A is not of the first Baire category, then A is of the second Baire category. A complement of A of the first BC, is called residual. BC thm says nonempty open subsets of a complete m.s is of the 2nd BC.

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A measurable function $f: E \rightarrow \mathbb{R}$ is essentially bounded by $M > 0$ if $|f(x)| \leq M$ for a.e $x \in E$, M is called an **essential upper bound**. We can define a norm on L_p like such

$$\|f\|_{\infty} = \text{ess sup}_{x \in E} |f(x)| = \inf_{M > 0} \{ |f(x)| < M \text{ a.e } x \in E \}$$

Proposition 23. for $1 \leq p \leq +\infty$, $\|f\|_p$ is a norm on $\{f: E \rightarrow \mathbb{R}: \int_E |f(x)|^p < \infty\} = L^p(E)$, and when $p = \infty$ $\{f: E \rightarrow \mathbb{R}: \|f\|_{\infty} < \infty\} = L^{\infty}(E)$, f here is always measurable.

Proposition 24 (Minkowski's Inequality). If $f, g \in L^p(E)$ then $\|f + g\|_p \leq \|f\|_p + \|g\|_p$

Proposition 25 (Holder's inequality). $\int_E |f(x)g(x)|dx \leq \|f\|_p \|g\|_q$ provided $\frac{1}{p} + \frac{1}{q} = 1$

Proposition 26 (Young's inequality). Suppose $\frac{1}{p} + \frac{1}{q} = 1$, then we have that for any $a, b \geq 0$, $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$

Proposition 27 (Holder's inequality part 2). suppose $f \neq 0$, then define

$$f^*(x) = \|f\|_p^{1-p} \cdot \operatorname{sgn} f(x) - \|f(x)\|^{p-1} \in L^p(E)$$

where

$$\operatorname{sgn}(a) = \begin{cases} 1 & \text{if } a > 0 \\ 0 & \text{if } a < 0 \end{cases}$$

and

$$\int_E f(x) f^*(x) dx = \|f\|_p \text{ note that } \|f^*\|_q = 1$$

Theorem 28 (Riez-Fischer). $L^p(E)$ is complete

Corollary 29. $C([a, b])$ is complete with respect to $\|f\|_\infty$ but is not complete with respect to $\|f\|_p$

Corollary 30. $L^p(E)$ is a Banach space, so it's a complete normed vector space.

Definition 31 (Rapidly Cauchy). Let $(X, \|\cdot\|)$ be a normed vector space, $\{f_n\}_{n \in \mathbb{N}}$ a sequence in X , $\{f_n\}_n$ is rapidly Cauchy if $\exists \{\varepsilon_n\}_n$ where $\varepsilon_n \geq 0$ $\forall n$ such that

$$\sum_{k=1}^{\infty} \varepsilon_k < +\infty$$

and

$$\|f_{k+1} - f_k\| \leq \varepsilon_k^2$$

An immediate observation is that if $\{f_n\}_n$ is rapidly Cauchy, then, using the triangle inequality for norms

$$\|f_{n+k} - f_n\| \leq \varepsilon_n^2 + \varepsilon_{n+1}^2 + \dots + \varepsilon_{n+k-1}^2$$

Theorem 32. let $E \subseteq \mathbb{R}^n$ be a measurable subset of $\mathbb{R}^n$, $1 \leq p \leq +\infty$, suppose $\{f_n\}_n$ is rapidly Cauchy in $L^p(E)$, then $f_n \rightarrow g \in L^p(E)$ in $L^p(E)$ norm, that is $\|f_n - g\| \rightarrow 0$ as $n \rightarrow \infty$, and pointwise $(f_n(x) \rightarrow g(x))$ for a.e $x \in E$.

Proof of theorem 32. Want to show pointwise first. This is a bit complicated, first of all by assumption, there exists $\{\varepsilon_k\}_k \geq 0$ such that $\sum_{k \geq 1} \varepsilon_k < +\infty$ and that $\|f_{k+1} - f_k\| < \varepsilon_k^2$, which means

$$\int_E |f_{k+1} - f_k|^p < \varepsilon_k^{2p}$$

Now observe that

$$m(\{x \in E : |f_{k+1}(x) - f_k(x)| \geq \varepsilon_k\}) = m(\{x \in E : |f_{k+1}(x) - f_k(x)|^p \geq \varepsilon_k^p\})$$

Since, for $p \geq 1$

$$\{x \in E: |f_{k+1}(x) - f_k(x)| \geq \varepsilon_k\} = \{x \in E: |f_{k+1}(x) - f_k(x)|^p \geq \varepsilon_k^p\}$$

Call $E_k = \{x \in E: |f_{k+1} - f_k| \geq \varepsilon_k\}$, then

$$m(E_k) \leq \frac{\int_E |f_{k+1} - f_k|^p}{\varepsilon_k^p} < \frac{\varepsilon_k^{2p}}{\varepsilon_k^p} = \varepsilon_k^p$$

We immediately get that, since $\sum_{k=1}^{\infty} \varepsilon_k < \infty$, it's necessary that for k large enough all $\varepsilon_k < 1$, that means, since $p \geq 1$, then

$$\sum_{k=1}^{\infty} \varepsilon_k^p < +\infty \implies \sum_{k=1}^{\infty} m(E_k) < +\infty$$

Then by Borel-Cantelli, we have that $m(\limsup_k E_k) = 0$ that means,

$$m(\{x \in E: x \in E_k \text{ infinitely many times}\}) = 0$$

which says for almost every $x \in E$, $\exists K(x) \in \mathbb{N}$ such that $\forall k \geq K(x)$, $x \notin E_k \iff |f_{k+1}(x) - f_k(x)| < \varepsilon_k$, and since $\varepsilon_k \rightarrow 0$, then this difference gets arbitrarily small, and if $n, k \geq K(x)$, then

$$|f_{n+k} - f_n| < \varepsilon_n + \varepsilon_{n+1} + \dots + \varepsilon_{n+k-1}$$

and this sum get's arbitrarily small as well, for n, k large enough (can group up subsequence ε_k 's and these will get arbitrarily small) so $\{f_n(x)\}$ is Cauchy for a.e $x \in E$, thus pointwise convergent to some f for a.e $x \in E$. This takes care of pointwise convergent, now we need to take care of convergence in $L^p(E)$. Notice that for any n, k , we have that

$$\|f_{n+k} - f_n\|_p \leq \varepsilon_n^2 + \varepsilon_{n+1}^2 + \dots + \varepsilon_{n+k-1}^2 \leq \sum_{j=n}^{\infty} \varepsilon_j^2$$

as n gets large and $k \geq n$. That means

$$\int_E |f_{n+k} - f_n|^p \leq \left(\sum_{j=n}^{\infty} \varepsilon_j^2 \right)^p$$

Now let $k \rightarrow \infty$, then observe that, By Fatou's lemma

$$\int_E |f - f_n|^p \leq \liminf_{k \rightarrow \infty} \int_E |f_{n+k} - f_n|^p \leq \left(\sum_{j=n}^{\infty} \varepsilon_j^2 \right)^p \rightarrow 0$$

Since the last inequality is preserved under limits and it's the tail of a convergent series, then

$$\|f - f_n\|_p \leq \sum_{j=n}^{\infty} \varepsilon_j^2 \rightarrow 0$$

so it's convergent in $L^p(E)$, now just need to verify if f is in $L^p(E)$, observe that

$$\|f\|_p \leq \|f_n - f\| + \|f_n - 0\| < +\infty$$

so $f \in L^p(E)$. □

Knowing this the proof of Riez-Fischer is easy,

Proof of theorem 28. Every Cauchy sequence has a rapidly cauchy subsequence and use previous theorem to conclude . □

Theorem 33. Let $E \subseteq \mathbb{R}^n$ be measurable, $1 \leq p \leq +\infty$, suppose $f_n(x) \in L^p(E) \rightarrow f(x)$ pointwise a.e on E and $f \in L^p(E)$, then

$$\|f_n - f\|_p \rightarrow 0 \iff \lim_{n \rightarrow \infty} \int_E |f_n|^p = \int_E |f|^p$$

where the left hand side means convergence in $L^p(E)$.

Proof. Royden and Fitzpatrick ch7, thm 7 □

A subset A of a metric space X is dense in X if and only if $\forall x \in X, \forall \varepsilon > 0, \exists y \in A$ such that $d(x, y) < \varepsilon$. If X is a normed space, $d(x, y) = \|x - y\|$, same definition just $d(x, y) = \|x - y\| < \varepsilon$.

A metric space X is separable if X has a countable dense subset.

Proposition 34. Let $E \subseteq \mathbb{R}^n$ measurable, $1 \leq p \leq +\infty$. Then $\{\text{simple functions in } L^p(E)\}$ is dense in $L^p(E)$. Simple functions are of the form $\phi = \sum_{k=1}^n c_k \cdot \chi_{E_k}$, where all the E_k 's are measurable and disjoint.

Proof. Follows directly from simple approximation lemma, if $f: E \rightarrow \mathbb{R}$ measurable, and $|f(x)| \leq M \geq 0$, then $\forall \varepsilon > 0, \exists \phi_E, \psi_E$, simple functions such that $\phi_E \leq f \leq \psi_E$, and $\psi_E - \phi_E < \varepsilon$ on E . It immediately follows that simple functions are dense in L^∞ , since the norm is just the supremum value of a function, and all functions in L^∞ have finite supremum, thus uniformly bounded by some $M \geq 0$. So for any function in L^∞ we are able to find simple functions that approximate that function in L^∞ an epsilon away. As for general L^p space, we need to make use of simple approximation theorem, where for all measurable functions $g: E \rightarrow \mathbb{R}^d$, there exists a sequence of simple functions $\{\varphi_n\}_n$ such that $\varphi_n(x) \rightarrow g(x)$ pointwise in E , and $|\varphi_n(x)| \leq |g(x)|$ on $E \forall n$, thus

$$\int_E |\varphi_n|^p \leq \int_E |g|^p < +\infty$$

So $\varphi_n \in L^p$ for all n . **Claim:** $\varphi_n \rightarrow g$ in $L^p(E)$ Proof of claim :

$$\begin{aligned} |\varphi_n - g|^p &\leq |\varphi_n + g|^p \\ &= 2^p \left| \frac{\varphi_n}{2} + \frac{g}{2} \right|^p \\ &\leq 2^{p-1} (|\varphi_n|^p + |g|^p) \quad \text{convexity of } |\cdot|^p \\ &\leq 2^p (|\varphi_n|^p + |g|^p) \leq 2^{p+1} |g|^p \text{ on } E \end{aligned}$$

We have that $\int_E 2^{p+1} |g|^p = 2^{p+1} \int_E |g|^p < \infty$ since $g \in L^p(E)$, and that is $\geq |\varphi_n - g|^p$ for all n , and therefore by dominated convergence theorem, since $|\varphi_n - g|^p \rightarrow 0$ as $n \rightarrow \infty$ for a.e $x \in E$, then

$$\int_E |\varphi_n - g|^p \rightarrow 0$$

as $n \rightarrow \infty$ which shows that every $g \in L^p$ is either a simple function or the convergent limit of a sequence of simple functions in $L^p(E)$, which shows that simple functions are dense in $L^p(E)$. □

Proposition 35. Let $[a, b] \subseteq \mathbb{R}$ closed and bounded interval, $1 \leq p < +\infty$, then step functions on $[a, b]$ are dense in $L^p([a, b])$

The proof can be found in Royden and Fitzpatrick ch7.4 prop 10

Proposition 36. $C_c(E)$ which is the set of continuous function with compact support on E are also dense in $L^p(E)$ with $1 \leq p < +\infty$

Theorem 37. Let E be a measurable subset of $\mathbb{R}^n$, and $1 \leq p < +\infty$, then $L^p(E)$ is separable (contains a countable dense subset)

Proof. Take step functions

$$\sum_{i=1}^n c_i \chi_{[-a_i, b_i]}$$

where $a_i, b_i, c_i \in \mathbb{R}$ □

Note that $L^\infty(E)$ is not separable

Definition 38 (Linear functional). Let X be a vector space over $\mathbb{R}$, then $f: X \rightarrow \mathbb{R}$ is a linear function if $f(ax + by) = af(x) + bf(y) \forall a, b \in \mathbb{R}, \forall x, y \in X$. Note $f(0) = 0$

These are linear transformations, and in a finite dimensional vector space, these have a matrix form (recall from linear algebra)

Definition 39 (bounded linear map). suppose X and Y are normed vector spaces, then a linear map $F: X \rightarrow Y$ is bounded, if and only if $\exists M > 0$ such that

$$\|F(x)\|_Y \leq M \cdot \|x\|_X$$

An important reminder is that if $x \neq \vec{0}$, then

$$\frac{\|F(x)\|_Y}{\|x\|_X}$$

is invariant if we replace x by tx , $t \in \mathbb{R}$, where $t \neq 0$. If F is bounded, we can define an operator norm of F by

$$\|F\|_{op} = \sup_{x \neq 0: x \in X} \frac{\|F(x)\|_Y}{\|x\|_X}$$

Proposition 40. Let $F: X \rightarrow Y$ linear map/operator from X to Y , then F bounded $\iff F$ continuous $\iff F$ continuous at $x = \vec{0} \in X$

Proposition 41. Bounded linear operators from X to Y form a vector space with $\|F_1 + F_2\|_{op} \leq \|F\|_{op} + \|F_2\|_{op}$

Proof. Indeed, consider

$$\sup_{x \neq 0} \frac{\|F_1(x) + F_2(x)\|_Y}{\|x\|_X} \leq \sup_{x \neq 0} \frac{\|F_1(x)\|_Y + \|F_2(x)\|_Y}{\|x\|_X} \leq \|F_1\|_{op} + \|F_2\|_{op}$$

□

To compute $\|F\|_{op}$ simply need to compute the supremum for $\{x \in X: \|x\|_X = 1\}$ since invariant under rescaling. Indeed if you consider the supremum norm, which is

$$\|F\|_{op} = \sup_{x \in X: x \neq 0} \frac{\|F(x)\|_Y}{\|x\|_X} = \sup_{x \in X: x \neq 0} \frac{\|F\left(\frac{x}{\|x\|_X}\right)\|_Y}{\left\|\frac{x}{\|x\|_X}\right\|_X} = \sup_{x \in X, \|x\|=1} \frac{\|F(x)\|_Y}{\|x\|_X}$$

Additionally note that if $F: X \rightarrow Y$ and $G: Y \rightarrow Z$ are bounded linear operators, then $G \circ F: X \rightarrow Z$ are also bounded linear operators.

this also defines the triangle inequality with the operator norm, we can verify that

$$\|G \circ F\|_{op} \leq \|G\|_{op} \cdot \|F\|_{op}$$

Bounded linear operator is also a bounded linear map $F: X \rightarrow Y$. An example would be that for $1 \leq p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$ and $f \in L^p(E)$, $g \in L^q(E)$, then

$$\Phi_g(f) = \int_E f(x)g(x)dx$$

is also a linear map. Indeed, we have holders inequality that gives us

$$\left| \int_E fg \right| \leq \int_E |fg| \leq \|f\|_p \|g\|_q$$

so we have that

$$\|\Phi\|_{op} = \sup_{f \neq 0} \frac{\left| \int_E fg \right|}{\|f\|_p} \leq \|g\|_q$$

And using part 2 of Holder's inequality

$$\exists f^* \in L^q(E): \|f^*\|_q = 1, \|f\|_p = \int_E f \cdot f^*, \|g\|_q = \int_E g^* \cdot g: \|g^*\|_p = 1$$

Claim: $\|\Phi_g\| = \|g\|_q$, take $f = g^*$, then $\frac{\Phi_g(g^*)}{\|g^*\|_p} = \int_E g g^* = \|g\|_q = \|g\|_q$

Theorem 42 (Riesz representation theorem). For $1 \leq p < +\infty$, any bounded linear functional on $L^p(E)$ has the form Φ_g for some $g \in L^p(E)$.

Lemma 43. Let E be a measurable subset of $\mathbb{R}$, $1 \leq p < +\infty$, g integrable function on E such that $\exists M > 0$ such that for any simple function $f \in L^p(E)$ we have $|\int_E g f| \leq M \|f\|_p$. Then $g \in L^q(E)$ and $\|g\|_q \leq M$.

Proof. This is hella simple. First step, simple approximation theorem, we know that there exists a sequence of non-negative simple functions $\{\varphi_n\}_n$ monotonically increasing and pointwise converging to g such that $\varphi_n \leq |g|$ for all n . We want to show that $\int_E \varphi_n^q \leq M^q$ since we can directly conclude by Fatou's that

$$\int_E |g|^q \leq \liminf_{n \rightarrow \infty} \int_E \varphi_n^q \leq M^q$$

where the conclusion follows immediately. To show that claim, observe that

$$\varphi_n^q = \varphi_n \cdot \varphi_n^{q-1} \leq |g| \cdot \varphi_n^{q-1} = g \cdot \operatorname{sgn}(g) \cdot \varphi_n^{q-1}$$

Letting $f_n = \operatorname{sgn}(g) \cdot \varphi_n^{q-1}$, then observing that f_n is simple with finite support, we have that, by assumption,

$$\int_E \varphi_n^q \leq \left| \int_E g f_n \right| \leq M \|f_n\|_p$$

but we have that $\int_E |f_n|^p = \int_E |f_n|^p = \int_E (\varphi_n^{q-1})^p = \int_E \varphi_n^q$, thus

$$\int_E \varphi_n^q \leq M \left(\int_E \varphi_n^q \right)^{\frac{1}{p}}$$

which gives us

$$M \geq \left(\int_E \varphi_n^q \right)^{1 - \frac{1}{p}} = \left(\int_E \varphi_n^q \right)^{\frac{1}{q}} = \|\varphi_n\|_q$$

and the result follows. $\square$

Theorem 44 (Riesz representation for closed intervals in $\mathbb{R}$). Let $[a, b] \subseteq \mathbb{R}$ closed and bounded, T a bounded linear functional on $L^p([a, b])$, $1 \leq p < +\infty$. Then $\exists g \in L^q([a, b])$ such that $T(f) = \int_a^b g \cdot f \forall f \in L^p([a, b])$.

Proof. This isn't all that more complicated, let T be a bounded linear functional on $L^p([a, b])$. Then we define $\Phi(x) = T(\chi_{[a, b]})$. We claim that this is absolutely continuous. That means, $\forall \varepsilon > 0, \exists \delta > 0$ such that for any finite sequence of disjoint intervals, $\{[a_k, b_k]\}_{k=1}^n$ where $\sum_{k=1}^n |b_k - a_k| < \delta$ then $\sum_{k=1}^n |\Phi(b_k) - \Phi(a_k)| < \varepsilon$. We need to check this, first of all, let $[c, d] \subseteq [a, b]$, notice that, by linearity

$$\Phi(d) - \Phi(c) = T(\chi_{[a, d]}) - T(\chi_{[a, c]}) = T(\chi_{[c, d]})$$

that means letting $\{[a_k, b_k]\}_{k=1}^n$ be a finite sequence of disjoint intervals all subsets of $[a, b]$, with $\sum_{k=1}^n |b_k - a_k| < \delta$, then we have that

$$\begin{aligned} \sum_{k=1}^n |\Phi(b_k) - \Phi(a_k)| &= \sum_{k=1}^n \varepsilon_k (\Phi(b_k) - \Phi(a_k)) \quad \text{where } \varepsilon_k = \text{sgn}(\Phi(b_k) - \Phi(a_k)) \\ &= \sum_{k=1}^n \varepsilon_k T(\chi_{[a_k, b_k]}) \\ &= T\left(\sum_{k=1}^n \chi_{[a_k, b_k]} \varepsilon_k\right) \quad \text{linearity} \\ &= \|T\|_{op} \left(\sum_{k=1}^n (b_k - a_k)\right)^{\frac{1}{p}} < \varepsilon \end{aligned}$$

where, in the last line, we choose $\delta < \left(\frac{\varepsilon}{\|T\|_{op}}\right)^p$. A claim that we just throw out there is that an absolutely continuous function on $[a, b]$ is differentiable on $[a, b]$ and Φ' exists in $[a, b]$, thus letting $g = \Phi'$, knowing that $\Phi(x) = \int_a^x \Phi'(x) dx$, we now want to show that correct form for T . Knowing that $T(\chi_{[c, d]}) = \Phi(d) - \Phi(c) = \int_a^b g \chi_{[c, d]}$, since all step functions on $[a, b]$ are a linear combination of coefficients times indicator functions for pairwise disjoint intervals of the form $[c, d]$, then by linearity, we have $T(\phi) = \int_a^b g \phi$ for any step function ϕ . Since step functions are dense in simple functions in $L^p([a, b])$, then we have that for any simple function f there exists a sequence of step functions $\{\varphi_n\}_n$ such that $\varphi_n \leq f$ for all n such that $\varphi_n \rightarrow f$ pointwise and in $L^p(E)$. By continuity of bounded linear operators, we have that $T(\varphi_n) \rightarrow T(f)$. By lebesgue dominated convergence theorem, we have that $\int_a^b g \varphi_n \rightarrow \int_a^b g f$ and by the previous lemma, we have that $g \in L^q([a, b])$ and that $\|g\|_q \leq \|T\|_{op}$. $\square$

Now we go on to prove Riesz representation theorem for general measurable sets. We have that

Theorem 45 (Riesz representation theorem for general measurable sets). Let $E \subseteq \mathbb{R}$ measurable, $1 \leq p < +\infty$ and T is a bounded linear functional on $L^p(E)$ then $\exists! g \in L^q(E)$ such that $T(f) = \int_E f \cdot g$ and $\|T\|_* = \|g\|_q$

First of all we prove uniqueness, if g_1 and g_2 both in $L^q(E)$ satisfy the above conditions, then $\int_E g_1 f = \int_E g_2 f$ for all $f \in L^p(E)$, thus $\int_E f(g_1 - g_2) = 0$ for all f in $L^p(E)$. This means that $\|g_1 - g_2\|_q = \|T - T\|_* = 0$ x

Proof. Let $T: L^p(\mathbb{R}) \rightarrow \mathbb{R}$ be a bounded linear functional on $\mathbb{R}$, then that means we can define a bounded linear function on $L^p([-n, n])$ as such.

$$T_n(f) = T(f\chi_{[-n, n]})$$

Which is not exactly a restriction of $L^p(\mathbb{R})$ to $L^p([-n, n])$, it's more like a quotient. This makes sense, since every function in $L^p([-n, n])$ can be extended to have domain $\mathbb{R}$ by putting some 0s, every $f \in L^p([-n, n])$ has it's extension in $L^p(\mathbb{R})$. Just like every function in $f \in L^p(\mathbb{R})$ has it's restriction $f\chi_{[-n, n]} \in L^p([-n, n])$, therefore, we can think of T_n as a functional $L^p([-n, n]) \rightarrow \mathbb{R}$, so by the previous result $\exists g_n \in L^q([-n, n])$ such that

$$T(f\chi_{[-n, n]}) = T_n(f) = \int_{-n}^n g_n f$$

if $g_n \rightarrow g \forall x \in \mathbb{R}$ for some $g \in L^p(\mathbb{R})$, more specifically g_n can be extended to some $g: \mathbb{R} \rightarrow \mathbb{R}$, then taking limits on the equation above gets us what we want. We claim, without proof, that $g_{n+k}\chi_{[-n, n]} \equiv g_n$, so we are actually able to extend g_n to some $g: \mathbb{R} \rightarrow \mathbb{R}$, where $g_n = g\chi_{[-n, n]}$, thus $g_n \rightarrow g$, complete proof later, this shit don't make too much sense

□

We move on to Hilbert spaces, and inner product spaces. But first some reminders about inner product spaces.

Definition 46 (Inner product). Let V be a vector space over F . An inner product on V is a function that assigns to every ordered pair of vectors x and y in V , a scalar in F , denoted $\langle x, y \rangle$, such that for all x, y , and z in V , and all $c \in F$, the following hold

1. $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$
2. $\langle cx, y \rangle = c \langle x, y \rangle$
3. $\overline{\langle x, y \rangle} = \langle y, x \rangle$ where the bar denotes complex conjugation
4. $\langle x, x \rangle > 0$ if $x \neq 0$

Theorem 47 (Inner product space). Let V be an inner product space. Then for $x, y, z \in V$ and $c \in F$, the following statements are true.

- (a) $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$.
- (b) $\langle x, cy \rangle = \overline{c} \langle x, y \rangle$.
- (c) $\langle x, 0 \rangle = \langle 0, x \rangle = 0$.

(d) $\langle x, x \rangle = 0$ if and only if $x = 0$.

(e) If $\langle x, y \rangle = \langle x, z \rangle$ for all $x \in V$, then $y = z$.

Definition. Let V be an inner product space. For $x \in V$, we define the norm or length of x by $\|x\| = \sqrt{\langle x, x \rangle}$.

Theorem 48. Let V be an inner product space over F . Then for all $x, y \in V$ and $c \in F$, the following statements are true.

1. $\|cx\| = |c| \cdot \|x\|$.
2. $\|x\| = 0$ if and only if $x = 0$. In any case, $\|x\| \geq 0$.
3. (Cauchy-Schwarz Inequality) $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$.
4. (Triangle Inequality) $\|x + y\| \leq \|x\| + \|y\|$.

Definition 49. Let V be an inner product space. Vectors x and y in V are orthogonal (perpendicular) if $\langle x, y \rangle = 0$. A subset S of V is orthogonal if any two distinct vectors in S are orthogonal. A vector x in V is a unit vector if $\|x\| = 1$. Finally, a subset S of V is orthonormal if S is orthogonal and consists entirely of unit vectors.

Definition 50 (ON basis). let V be an inner product space. A subset of V is an orthonormal basis for V if it is an ordered basis that is orthonormal.

Note that through the Graham-Schmidt process, we are always able to construct an orthogonal set from a linearly independent set of vectors in such a way that both sets generate the same subspace. For finite dimensional vector spaces, there always exists an orthonormal basis. For general Hilbert spaces that are not finite dimensional, things gets more complicated , but we'll prove some surprising theorems.

Definition 51 (Hilbert space). A Hilbert space is a complete inner product space.

Some examples include, ℓ^2 or $L^2(E)$. For both, we have the Cauchy-Schwartz inequality (Special case of Holder's inequality for $p = 2$). We can define the inner product on ℓ^2 as

$$\langle x, y \rangle = \sum_{i=1}^{\infty} x_i y_i$$

and the inner product on $L^2(E)$ as

$$\langle f, g \rangle = \int_E f(x) \overline{g(x)} dx$$

where

$$\|f\| = \left(\int_E |f(x)|^2 dx \right)^{\frac{1}{2}}$$

where the norm is just define to be $\|x\| = \sqrt{\langle x, x \rangle}$. Can easily check that the triangle inequality is also satisfied.

Proposition 52 (Triangle inequality and Cauchy Schwartz inequality). These inequalities have to hold in Any hilbert space

1. (Triangle inequality) $\|f + g\| \leq \|f\| + \|g\|$
2. (Cauchy Schwartz inequality) $|\langle f, g \rangle| \leq \|f\| \|g\|$

Theorem 53 (Parallelogram law). This completely defines an inner product space, where $\langle f + g, f + g \rangle + \langle f - g, f - g \rangle = 2(\|f\|^2 + \|g\|^2)$?

Proof. notice that

$$\langle f + g, f + g \rangle + \langle f - g, f - g \rangle = 2(\|f\|^2 + \|g\|^2)$$

You can easily verify this computation using pen and paper $\square$

Notice that by the Riezs representation theorem for general measurable sets we get that any bounded linear functional $T: L^2 \rightarrow L^2$ is given by an inner product with a function in L^2 since $T(f) = \int_E f \cdot g = \langle f, g \rangle$ for some $g \in L^2(E)$ such that $\|T\|_{op} = \|g\|_2$.

Definition 54 (Orthonormal sets). if $\langle f, g \rangle = 0$, where we write $f \perp g$ then $\|f + g\|^2 = \|f\|^2 + \|g\|^2$ (Pythagoras theorem)

Definition 55 (Orthonormal set). A set is orthonormal if $\langle e_i, e_j \rangle = \delta_{ij}$

Also, if $\{e_i\}_{i=1}^n$ is orthonormal, then $\|x_1 e_1 + \dots + x_n e_n\|^2 = \|x_1\|^2 + \dots + \|x_n\|^2$, this is just by expanding the inner product and cancelling out some terms.

Proposition 56. if $\{e_k\}_{k=1}^\infty$ is orthonormal, and $f = \sum_k a_k e_k \in \mathcal{H}$ where the sum is finitethen

$$\|f\|^2 = \sum_n |a_k|^2$$

This is straightforward application of Pythagoras theorem and using the fact that $\{e_k\}_{k=1}^\infty$ is an orthonormal set, so $\langle e_i, e_i \rangle = 1$ for any i .

Proposition 57. $\{e^{inx}\}$ is orthonormal in $L^2([-\pi, \pi])$

Proposition 58 (Fourier coefficients). We can define the inner product for $f, g \in L^2([-\pi, \pi])$ as $\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)} dx$, then we get that the n -th fourier coefficient of f is

$$\hat{f}(n) = \langle f, e^{inx} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx$$

Proposition 59. If $\sum_{n=-\infty}^{+\infty} |\hat{f}(n)|^2 < \infty$ then we get that

$$\left\| \sum_{n=-\infty}^{+\infty} \hat{f}(n) e^{inx} \right\|_2^2 = \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2$$

Remarks 60. We begin with some general remarks about Hilbert spaces. First of all if $\{e_i\}_{i=1}^\infty$ is orthonormal, then

$$\|f\|_2^2 = \sum_{i=1}^{\infty} \langle f, e_i \rangle^2$$

Furthermore, TFAE:

1. $\{e_i\}_{i=1}^\infty$ forms a basis of H
2. if $\langle f, e_i \rangle = 0 \forall i$, then $f = 0$
3. Let $a_i = \langle f, e_i \rangle$ and $S_n = \sum_{i=1}^n a_i e_i$, then $\|f - S_n(f)\|_2^2 \rightarrow 0$ as $n \rightarrow \infty$
4. Defining

$$f \rightarrow \sum_{i=1}^n \langle f, e_i \rangle e_i = F_n(f)$$

as the projection of f onto the space $\{e_1, \dots, e_n\}$, then $\forall f \in H$, $\|f\|_2^2 = \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 = \sum_{i=1}^{\infty} |a_i|^2$

Also finite linear combinations of $\{e_i\}_{i=1}^\infty$ are dense in H .

Remark 61. A fact that will be useful for the homework is that given $f \in L^1(-\pi, \pi)$, then the n -th Fourier coefficient of f is $\hat{f}(n) = \langle f(x), e^{inx} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$ where $n \in \mathbb{Z}$.

Proposition 62 (Criteria for Orthogonal projection). P is an orthogonal projection on a Hilbert space if and only if P is idempotent, that is $P^2 = P$ and self adjoint, that means $P^* = P$ to verify this we just need to verify that, $\forall x, y \in \mathcal{H}$

$$\langle Px, y \rangle = \langle x, Py \rangle$$

Proposition 63 (Some basic facts about inner products). First of all we have that if P is a orthogonal projection on a hilbert space $\mathcal{H}$, then we have these basics facts, $\forall x, y \in \mathcal{H}$

1. $\langle P(x), y - P(y) \rangle = 0$
2. $\langle (x - P(x)), y \rangle = 0$
3. $\langle x, Py \rangle = \langle Px, Py \rangle = \langle Px, y \rangle$

Proof. 1 $\implies$ 2, since finite linear combinations of $\{e_i\}_i$ are dense in $\mathcal{H}$, then we get that there is a sequence $\{g_n\}_n$ where each element is a finite linear combination of $\{e_i\}_i$ such that $\|f - g\| \rightarrow 0$ as $n \rightarrow \infty$, and we know that $\langle f, g_n \rangle = 0$ for all n , then a straightforward application of Cauchy-Schwartz gets us

$$\|f\|^2 = \langle f, f \rangle = \langle f, f \rangle - \langle f, g_n \rangle = \langle f, f - g_n \rangle \leq \|f\| \|f - g_n\| \rightarrow 0$$

as n tends towards ∞

2 $\implies$ 3, is given by observing that, first of all

□

Theorem 64 (Hilbert space basis existence). Every Hilbert space has an orthonormal basis

The proof for this uses Zorn's lemma (axiom of choice)

Lemma 65 (Zorns). Every non-empty partially ordered set in which every chain is bounded also contains a maximal element

Definition 66 (Partially ordered set). $(X, \leq)$ is a partially ordered set if

- $x \leq x \ \forall x \in X$
- if $x, y \in X$ with $x \leq y$ and $y \leq x$, then $x = y$
- if $x, y, z \in X$, with $x \leq y$, $y \leq z$, then $x \leq z$

Definition 67 (Chain). A chain A is a totally ordered subset of X where $\forall x, y \in A$, either $x \leq y$ or $y \leq x$.

Definition 68 (Bounded chain). $Y \subseteq X$ is bounded if $\exists z \in X$ such that $\forall y \in Y$, $y \leq z$.

Definition 69 (maximal element). A maximal element m of X is an element where if $m \leq x$ for $x \in X$, then $x = m$.

We want to use Zorn's lemma to prove

Proof of Theorem 63. Let H be a Hilbert space, and let $\mathcal{O}$ be the collection of all orthonormal subsets of H . $\mathcal{O}$ is non empty because $\{x\} \in \mathcal{O}$ where x is just any element of H with $\|x\| = 1$. We can define a partial ordering by set inclusion. Observing that every chain $\mathcal{A} \subseteq \mathcal{O}$ has an upper bound, by taking the union of every set in $\mathcal{A}$. Knowing this Zorn's lemma guarantees the existence of a maximal element, call it B , we want to show that this forms a basis of H . It's enough, as shown previously, to show that the linear span of B is dense in H , and this is if and only if the closure of $\text{span}B$, denoted, $\overline{\text{span}B}$ ends up being the whole space.

Remark 70 (Orthogonal decomposition theorem). Letting H be a hilbert space and M be a closed subset of H , define the orthogonal complement of M as

$$M^\perp = \{y \in X : \langle x, y \rangle = 0, x \in X\}$$

then for any $x \in H$, $\exists!$ decomposition where

$$x = Px + Qx$$

where $Px \in M$ and $Qx \in M^\perp$, where P and Q are the projection of x to M and $M^\perp$, respectively. Generally we have that $H = M \oplus M^\perp$

Thus we can take the closure of the linear span of B , $\overline{\text{span}B}$, by the orthogonal decomposition theorem, if $\text{span}B \neq H$, then

$$H = \overline{\text{span}B} + \overline{\text{span}B}^\perp$$

But that means exists some $y \in \overline{\text{span}B}^\perp$ with $\|y\| = 1$ that is orthonormal to every element in B , since $B \subseteq \overline{\text{span}B}$, thus $B \cup \{y\}$ is a larger orthonormal subset, contradicting maximality of B , and we are done. $\square$

Definition 71 (Unitary maps). Unitary maps are linear maps that preserve the inner product. Formally, a mapping $U: \mathcal{H} \rightarrow \mathcal{H}'$ is called unitary if

1. U is linear that is $U(\alpha f + \beta g) = \alpha U(f) + \beta U(g)$
2. U is a bijection
3. $\|Uf\|_{\mathcal{H}'} = \|f\|_{\mathcal{H}}$ for all $f \in \mathcal{H}$.

We are able to make some observations. First, since U is bijective, then it has an inverse $U': \mathcal{H}' \rightarrow \mathcal{H}$. (3) also implies that if U is unitary, then we have

$$\langle Uf, Ug \rangle_{\mathcal{H}'} = \langle f, g \rangle_{\mathcal{H}}$$

Proposition 72 (Unitary equivalent). Two Hilbert spaces are Unitarily equivalent if there exists a Unitary map between them. In fact, any two infinite-dimensional Hilbert spaces are unitarily equivalent. That mean

Proof. Consider two Hilbert spaces, $\mathcal{H}$ and $\mathcal{H}'$ with orthonormal basis $\{e_i\}_i$ and $\{e'_i\}_i$ respectively, then consider the mapping

$$U(f) = U\left(\sum_k a_k e_k\right) = \sum_k a_k e'_k$$

This is clearly linear and also bijective, and very obviously

$$\|U(f)\|_{\mathcal{H}'}^2 = \sum_k |a_k|^2 = \|f\|_{\mathcal{H}}^2$$

A direct consequence of Parseval's identity. $\square$

Definition 73 (Trigonometric Polynomials). Trigonometric polynomials are polynomials of the form

$$\sum_{n=-N}^{+N} a_n e^{inx} \quad (a_n \in \mathbb{C})$$

Proposition 74. Trigonometric polynomials are dense in $L^2([-\pi, \pi])$

Recalling that $C([-\pi, \pi])$ is dense in $L^2([-\pi, \pi])$. It suffices to show that for all $f \in C([-\pi, \pi])$, then $\forall \varepsilon > 0 \exists P$ a trig polynomial such that $\|f - P\|_2 < \varepsilon$. In fact we can show that $\exists P$ such that $\|f - P\|_\infty < \varepsilon$, which is even a stronger result.

Theorem 75 (Stone-Waierstrass). Let X be a compact, Hausdorff topological space. Let $\mathcal{A}$ be an algebra of continuous functions on X that separates points and contains constant functions. Then $\mathcal{A}$ is dense in $C(X)$.

Before any proofs, we define an algebra

Definition 76 (Algebra (not set theoretic)). define $\mathcal{A}$ to be an algebra, which is a set with multiplication addition and scalar multiplication on a field $\mathbb{K}$ such that $f, g \in \mathcal{A}$, and $c_1, c_2 \in \mathbb{K} \implies c_1f + c_2g \in \mathcal{A}$, and $f \cdot g \in \mathcal{A}$. (Examples include polynomials, trigonometric polynomials e^{inx})

Definition 77 (Separates points). We say a collection $\mathcal{F}$ of functions on X separates points if $\forall x, y \in X, x \neq y, \exists f \in \mathcal{F}$ such that $f(x) \neq f(y)$.

We ain't gonna give a full proof, but for now consider a more restricted result.

Theorem 78. $\forall f \in C([- \pi, \pi]), \forall \varepsilon > 0, \exists P$ a trigonometric polynomial such that $\|f - P\|_\infty < \varepsilon$.

Proof. Let $Q_1(x), Q_2(x), \dots$ be trigonometric polynomials such that

- $Q_k \geq 0 \ \forall k \in \mathbb{N} \ \forall x \in \mathbb{R}$
- $\frac{1}{2\pi} \int_{-\pi}^{\pi} Q_k(x) dx = 1 \ \forall k \in \mathbb{N}$
- Letting $\eta_k(\delta) = \sup_{\delta \leq |x| \leq \pi} Q_k(x)$. Then $\lim_{k \rightarrow \infty} \eta_k(\delta) = 0$

The higher level concept of the proof, is that given a sequence of trigonometric polynomials with these properties, given any function $f \in C([- \pi, \pi])$, we can construct a function, that is also a trigonometric polynomial such that for all $y \in [- \pi, \pi]$, these two are arbitrarily close. Notice that defining

$$P_k(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) Q_k(y) dy$$

. Using a change of variables and using the periodicity of e^{inx} we get that the latter integral is equal to

$$\begin{aligned} P_k(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) Q_k(x-y) dy = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \sum_{n=-N}^{+N} a_{n,k} e^{in(x-y)} dy \\ &= \sum_{n=-N}^{+N} a_{n,k} e^{inx} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} dy \\ &= \sum_{n=-N}^{+N} a_{n,k} \hat{f}_n(y) e^{inx} \end{aligned}$$

which it itself a trigonometric polynomial, and then we take any function $f \in C([- \pi, \pi])$. Knowing this, let $\varepsilon > 0$, since f is continuous on a compact interval,

then it's uniform continuous, meaning that $\forall x_1, x_2 \in [-\pi, \pi]$ if $|x_1 - x_2| < \delta$, then $|f(x_1) - f(x_2)| < \varepsilon$ and then we just need to evaluate

$$|f(y) - P_k(y)| = \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(y-x) - f(y))Q_k(y)dx = I_1 + I_2$$

, since $\int Q_k(x) = 1$, we can bring $f(y)$ into the integral, since it doesn't depend on x . where

$$I_1 = \frac{1}{2\pi} \int_{-\delta}^{\delta} (f(y-x) - f(y))Q_k(y)dx$$

and

$$I_2 = \frac{1}{2\pi} \int_{[-\pi, -\delta] \cup [\delta, \pi]} (f(y-x) - f(y))Q_k(y)dx$$

Observing that

$$\begin{aligned} I_1 &= \frac{1}{2\pi} \int_{-\delta}^{\delta} (f(y-x) - f(y))Q_k(y)dx \\ &\leq \frac{1}{2\pi} \int_{-\delta}^{\delta} |f(y-x) - f(y)|Q_k(y)dx \\ &\leq \frac{1}{2\pi} \int_{-\delta}^{\delta} \varepsilon Q_k(y)dx \leq \varepsilon \end{aligned}$$

Since $|(y-x) - y| = |x| \leq \delta$ in the case of I_1 Now we just need to take care of I_2 . Note that $\exists M \in \mathbb{N}$ such that $|f(y-x) - f(y)| \leq \sup_{a,b \in [-\pi, \pi]} |f(b) - f(a)| \leq 2M$ since f reaches it's maxima on a compact interval.

$$\begin{aligned} I_2 &= \frac{1}{2\pi} \int_{[-\pi, -\delta] \cup [\delta, \pi]} (f(y-x) - f(y))Q_k(y)dx \\ &\leq \frac{1}{2\pi} \int_{[-\pi, -\delta] \cup [\delta, \pi]} 2M\eta_k(\delta)dx = 2M\eta_k(\delta) \end{aligned}$$

which vanishes to 0 as $k \rightarrow \infty$, by the third property, thus proving that the difference vanishes, so we are done with the main part of the proof. Now we are left to construct $Q_1(x), Q_2(x), \dots$ with the properties as described above. We can just choose $Q_k(x) = c_k \left(\frac{1 + \cos x}{2} \right)^k$, where c_k is a constant such that the

integral turns out to be 1. and also observe that, given c_k chosen appropriately

$$\begin{aligned}
1 &= \frac{c_k}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1 + \cos x}{2} \right)^k dx \\
&= \frac{c_k}{\pi} \int_0^{\pi} \left(\frac{1 + \cos x}{2} \right)^k dx \quad \text{by symmetry} \\
&> \frac{c_k}{\pi} \int_0^{\pi} \left(\frac{1 + \cos x}{2} \right)^k \sin x \, dx \\
&= \frac{c_k}{n} \int_1^0 u^k (-2du) \\
&= \frac{c_k}{n} \int_0^1 u^k (2du) \\
&= \frac{2c_k}{\pi} \cdot \frac{1}{k+1}
\end{aligned}$$

Using the substitution $u = \frac{1+\cos x}{2}$ and $du = -\frac{1}{2} \sin x dx$. We then get that $c_k < \frac{\pi}{2}(k+1)$ so $Q_k(x) < \frac{\pi}{2}(k+1) \left(\frac{1+\cos x}{2} \right)^k \rightarrow 0$ for $x \neq 0$, so we recover property 3. $\square$

Corollary 79. trigonometric polynomials are dense in $L^2([-\pi, \pi])$

Proof. Trig polys are dense in $C([-\pi, \pi])$ under $\|\cdot\|_{\infty}$. If $|P_k(x) - f(x)| < \varepsilon$, then we simply get that

$$\|P_k - f\|_2 = \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x) - P_k(x)|^2 dx \right)^{\frac{1}{2}} < \sqrt{\varepsilon^2} = \varepsilon$$

$\square$

Theorem 80. Let X, Y be Banach spaces, $F: X \rightarrow Y$ bounded linear operator, and F is surjective onto Y . Let U be the open unit ball in X and V be the open unit ball. Then $\exists \delta > 0$ such that $F(U) \supset \delta V \implies F(B_r(x_0)) \supseteq B_{\delta r}(F(x_0))$

Proof. $\forall y \in Y, \exists x \in X$, such that $F(x) = y$. If $\|x\| \leq k$, then we have something $y \in F(k \cdot U)$, $Y = \bigcup_{k \in \mathbb{N}} F(k \cdot U)$, Y complete. By Baire Category theorem, $\exists k$ such that $\overline{F(k \cdot U)} \supseteq W$, for some W non empty open set in Y . For every $y \in W$, $\exists \{x_i\}_i \subseteq k \cdot U$ such that $F(x_i) \rightarrow y$, this is by the fact that y in the closure of $F(k \cdot U)$ and the fact that F is onto. So let k be sufficiently large such that $\overline{F(k \cdot U)}$ contains non empty interior, so open set W . Let $y_0 \in W$, $\exists \rho > 0$ such that $B(y_0, \rho) \subseteq W$, then $y_0 + y \in W$, if $\|y\| < \rho$, then $\exists \{x'_i\}_i, \{x''_i\}_i \subseteq k \cdot U$ such that $F(x'_i) \rightarrow y_0$, $F(x''_i) \rightarrow y_0 + y$, $F(x''_i - x'_i) \rightarrow y_0 + y - y_0 = y$. $\square$

Let $f \in L^1([- \Pi])$ where $\Pi = [-\pi, \pi]$. Recall Fourier coefficients of f are given by

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

We can also define a partial sum of the Fourier convolution

$$(S_n f)(x) = \sum_{k=-n}^n \hat{f}(k) e^{ikx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \underbrace{\left(\sum_{k=-n}^n e^{ikx} \right)}_{\text{Dirichlet Kernel } D_n(x)} dx$$